hvzomjveo

March 22, 2026

[479]:
```
import pandas as pd
import numpy as np
pd.set_option('display.max_columns',None)
```

[480]:
```
dfgames=pd.read_csv("games.csv")
dfgames2=pd.read_csv("games2.csv")
```

[481]:
```
dfgames.head()
```

[481]:
```
     AppID                                  Name  Release date  \
0  2539430             Black Dragon Mage Playtest   Aug 1, 2023
1   496350  Supipara - Chapter 1 Spring Has Come!  Jul 29, 2016
2  1034400      Mystery Solitaire The Black Raven   May 6, 2019
3  3292190                   - Vtuber Paranoia  Oct 31, 2024
4  3631080                          Maze Quest VR  Apr 24, 2025

  Estimated owners  Peak CCU  Required age  Price  DiscountDLC count  \
0            0 - 0         0             0   0.00                  0
1        0 - 20000         0             0   5.24                 65
2        0 - 20000         0             0   4.99                  0
3        0 - 20000         1             0   8.99                  0
4        0 - 20000         0             0   4.99                  0

   About the game                                Supported languages  \
0               0                                                NaN
1               0  Springtime, April: when the cherry trees come …
2               0  Immerse yourself in the most beloved, mystical…
3               1  synopsis 'Hello, I'm Hiyoro, a new YouTuber!' …
4               0  Its not just a Maze; its a Quest! Enter the ca…

                        Full audio languages      Reviews Header image  \
0                                         []           []          NaN
1                                ['English']           []          NaN
2  ['English', 'French', 'German', 'Russian']           []          NaN
3                                 ['Korean']   ['Korean']          NaN
4                                ['English']  ['English']          NaN
```

```
                                            Website  \
0  https://shared.akamai.steamstatic.com/store_it…
1  https://shared.akamai.steamstatic.com/store_it…
2  https://shared.akamai.steamstatic.com/store_it…
3  https://shared.akamai.steamstatic.com/store_it…
4  https://shared.akamai.steamstatic.com/store_it…

                                 Support url  \
0                                        NaN
1              http://mangagamer.org/supipara
2       https://www.facebook.com/8FloorGames/
3                                        NaN
4  https://www.realityexpanded.com/books-games

                         Support email                       Windows   Mac  \
0                                  NaN                           NaN  True
1                http://mangagamer.com       support@mangagamer.com  True
2  https://www.facebook.com/8FloorGames           support@8floor.net  True
3                                  NaN        yujingamesc@gmail.com  True
4      https://www.realityexpanded.com  support@realityexpanded.com  True

   Linux  Metacritic score  Metacritic url User score  Positive  Negative  \
0  False             False               0        NaN         0         0
1  False             False               0        NaN         0       252
2   True             False               0        NaN         0        21
3  False             False               0        NaN         0         0
4  False             False               0        NaN         0         0

   Score rank  Achievements  Recommendations  Notes  \
0           0           NaN                0      0
1           3           NaN                0    231
2           3           NaN                0      0
3           0           NaN               19      0
4           0           NaN                0      0

                            Average playtime forever  \
0                                                NaN
1                                                NaN
2                                                NaN
3  The game includes the following elements. 1. G…
4                                                NaN

   Average playtime two weeks  Median playtime forever  \
0                           0                        0
1                           8                        0
2                           0                        0
3                           0                        0
```

```
4                          0                       0

   Median playtime two weeks  Developers            Publishers  \
0                          0           0                   NaN
1                          8           0                minori
2                          0           0           Somer Games
3                          0           0
4                          0           0  Reality Expanded LLC

            Categories                                            Genres  \
0                  NaN                                               NaN
1           MangaGamer  Single-player,Steam Trading Cards,Steam Cloud,…
2               8floor                       Single-player,Family Sharing
3                      Single-player,Steam Achievements,Family Sharing
4  Reality Expanded LLC  Single-player,VR Only,Steam Leaderboards,Famil…

                     Tags                                       Screenshots  \
0                     NaN                                               NaN
1                Adventure                   Adventure,Visual Novel,Anime,Cute
2                   Casual  Casual,Card Game,Solitaire,Puzzle,Hidden Objec…
3  Casual,Indie,Simulation                                               NaN
4      Action,Early Access                                               NaN

                                            Movies  Column1
0  https://shared.akamai.steamstatic.com/store_it…      NaN
1  https://shared.akamai.steamstatic.com/store_it…      NaN
2  https://shared.akamai.steamstatic.com/store_it…      NaN
3  https://shared.akamai.steamstatic.com/store_it…      NaN
4  https://shared.akamai.steamstatic.com/store_it…      NaN
```

[482]:
```
dfgames.shape
```

[482]:
```
(122611, 40)
```

[483]:
```
dfgames2.head()
```

[483]:
```
           ID                              Tittle ReleaseDate Rating  \
0  1300001620                  Perfect Dark (2000)  2000-05-22      M
1  1300027043                  Grand Theft Auto IV  2008-04-29      M
2  1300003547             Tony Hawk's Pro Skater 3  2001-10-30      T
3  1300110124                   Grand Theft Auto V  2014-11-18      M
4  1300001290  The Legend of Zelda: Ocarina of Time  1998-11-23      E

              Genres                                       Description  \
0                FPS  [Xbox Live Arcade]  Agent Joanna Dark hit the …
1  Open-World Action  [Metacritic's 2008 PS3 Game of the Year; Also …
2            Skating  Challenge up to four friends in online competi…
```

```
3  Open-World Action  Los Santos is a vast, sun-soaked metropolis fu…
4  Open-World Action  As a young boy, Link is tricked by Ganondorf, …

                                          Platforms  Metascore  \
0                              Nintendo 64,Xbox 360       97.0
1                          PlayStation 3,Xbox 360,PC       98.0
2  Xbox,PlayStation 2,GameCube,PlayStation,PC,Gam…       97.0
3  PlayStation 3,Xbox 360,PlayStation 4,Xbox One,…       97.0
4                                       Nintendo 64       99.0

   Metascore_count Metascore_sentiment  Userscore  Userscore_count  \
0               30   Universal acclaim         84              664
1               86   Universal acclaim         83             6031
2               34   Universal acclaim         76              712
3               66   Universal acclaim         85            15292
4               22   Universal acclaim         91            12158

   Userscore_sentiment   Platform_metascores                Developer  \
0  Generally favorable                 97,79                Rare Ltd.
1  Generally favorable              98,98,90           Rockstar North
2  Generally favorable     93,97,91,87,90,90  Neversoft Entertainment
3  Generally favorable  97,97,97,97,96,81,79           Rockstar North
4   Universal acclaim                    99                 Nintendo

                              Publisher
0  Rare Ltd.,Xbox Game Studios,Nintendo
1                Rockstar Games,Capcom
2                   Activision,Success
3   Rockstar Games,Take-Two Interactive
4                   Nintendo,Gradiente
```

[484]:
```
dfgames.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 122611 entries, 0 to 122610
Data columns (total 40 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   AppID                       122611 non-null  int64
 1   Name                        122610 non-null  str
 2   Release date                122611 non-null  str
 3   Estimated owners            122611 non-null  str
 4   Peak CCU                    122611 non-null  int64
 5   Required age                122611 non-null  int64
 6   Price                       122611 non-null  float64
 7   DiscountDLC count           122611 non-null  int64
 8   About the game              122611 non-null  int64
```

```
 9   Supported languages         114162 non-null  str
 10  Full audio languages        122611 non-null  str
 11  Reviews                     122611 non-null  str
 12  Header image                12070 non-null   str
 13  Website                     122530 non-null  str
 14  Support url                 49676 non-null   str
 15  Support email               54142 non-null   str
 16  Windows                     100348 non-null  str
 17  Mac                         122611 non-null  bool
 18  Linux                       122611 non-null  bool
 19  Metacritic score            122611 non-null  bool
 20  Metacritic url              122611 non-null  int64
 21  User score                  4256 non-null    str
 22  Positive                    122611 non-null  int64
 23  Negative                    122611 non-null  int64
 24  Score rank                  122611 non-null  int64
 25  Achievements                40 non-null      float64
 26  Recommendations             122611 non-null  int64
 27  Notes                       122611 non-null  int64
 28  Average playtime forever    22458 non-null   str
 29  Average playtime two weeks  122611 non-null  int64
 30  Median playtime forever     122611 non-null  int64
 31  Median playtime two weeks   122611 non-null  int64
 32  Developers                  122611 non-null  int64
 33  Publishers                  114174 non-null  str
 34  Categories                  113702 non-null  str
 35  Genres                      113658 non-null  str
 36  Tags                        114198 non-null  str
 37  Screenshots                 83346 non-null   str
 38  Movies                      116593 non-null  str
 39  Column1                     0 non-null       float64
dtypes: bool(3), float64(3), int64(15), str(19)
memory usage: 35.0 MB
```

[485]:
```
dfgames.columns
```

[485]:
```
Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
       'Required age', 'Price', 'DiscountDLC count', 'About the game',
       'Supported languages', 'Full audio languages', 'Reviews',
       'Header image', 'Website', 'Support url', 'Support email', 'Windows',
       'Mac', 'Linux', 'Metacritic score', 'Metacritic url', 'User score',
       'Positive', 'Negative', 'Score rank', 'Achievements', 'Recommendations',
       'Notes', 'Average playtime forever', 'Average playtime two weeks',
       'Median playtime forever', 'Median playtime two weeks', 'Developers',
       'Publishers', 'Categories', 'Genres', 'Tags', 'Screenshots', 'Movies',
       'Column1'],
      dtype='str')
```

[486]:
```
dfgames2.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 13913 entries, 0 to 13912
Data columns (total 16 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   ID                    13913 non-null  int64
 1   Tittle                13913 non-null  str
 2   ReleaseDate           13884 non-null  str
 3   Rating                11610 non-null  str
 4   Genres                13913 non-null  str
 5   Description           13871 non-null  str
 6   Platforms             13912 non-null  str
 7   Metascore             13912 non-null  float64
 8   Metascore_count       13913 non-null  int64
 9   Metascore_sentiment   13912 non-null  str
 10  Userscore             13913 non-null  int64
 11  Userscore_count       13913 non-null  int64
 12  Userscore_sentiment   12440 non-null  str
 13  Platform_metascores   13912 non-null  str
 14  Developer             13904 non-null  str
 15  Publisher             13911 non-null  str
dtypes: float64(1), int64(4), str(11)
memory usage: 1.7 MB
```

[487]:
```
dfgames = dfgames.drop(columns=[
    'Column1',
    'Header image',
    'Website',
    'Support url',
    'Support email',
    'Screenshots',
    'Movies',
    'Metacritic url',
    'About the game',
    'Supported languages',
    'Full audio languages',
    'Notes',
    'Tags'
])
```

[488]:
```
dfgames.shape
```

[488]: (122611, 27)

[489]:
```
dfgames.columns
```

[489]:
```
Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
       'Required age', 'Price', 'DiscountDLC count', 'Reviews', 'Windows',
       'Mac', 'Linux', 'Metacritic score', 'User score', 'Positive',
       'Negative', 'Score rank', 'Achievements', 'Recommendations',
       'Average playtime forever', 'Average playtime two weeks',
       'Median playtime forever', 'Median playtime two weeks', 'Developers',
       'Publishers', 'Categories', 'Genres'],
      dtype='str')
```

[490]:
```
dfgames2.columns
```

[490]:
```
Index(['ID', 'Tittle', 'ReleaseDate', 'Rating', 'Genres', 'Description',
       'Platforms', 'Metascore', 'Metascore_count', 'Metascore_sentiment',
       'Userscore', 'Userscore_count', 'Userscore_sentiment',
       'Platform_metascores', 'Developer', 'Publisher'],
      dtype='str')
```

[491]:
```
dfgames2 = dfgames2.drop(columns=[
    'ID',
    'Description',
    'Platform_metascores',
    'Metascore_sentiment',
    'Userscore_sentiment',
    'Metascore_count',
    'Userscore_count'
])
```

[492]:
```
dfgames2.columns
```

[492]:
```
Index(['Tittle', 'ReleaseDate', 'Rating', 'Genres', 'Platforms', 'Metascore',
       'Userscore', 'Developer', 'Publisher'],
      dtype='str')
```

[493]:
```
dfgames.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 122611 entries, 0 to 122610
Data columns (total 27 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   AppID                       122611 non-null  int64
 1   Name                        122610 non-null  str
 2   Release date                122611 non-null  str
 3   Estimated owners            122611 non-null  str
 4   Peak CCU                    122611 non-null  int64
 5   Required age                122611 non-null  int64
 6   Price                       122611 non-null  float64
```

```
 7   DiscountDLC count           122611 non-null  int64
 8   Reviews                     122611 non-null  str
 9   Windows                     100348 non-null  str
 10  Mac                         122611 non-null  bool
 11  Linux                       122611 non-null  bool
 12  Metacritic score            122611 non-null  bool
 13  User score                  4256 non-null    str
 14  Positive                    122611 non-null  int64
 15  Negative                    122611 non-null  int64
 16  Score rank                  122611 non-null  int64
 17  Achievements                40 non-null      float64
 18  Recommendations             122611 non-null  int64
 19  Average playtime forever    22458 non-null   str
 20  Average playtime two weeks  122611 non-null  int64
 21  Median playtime forever     122611 non-null  int64
 22  Median playtime two weeks   122611 non-null  int64
 23  Developers                  122611 non-null  int64
 24  Publishers                  114174 non-null  str
 25  Categories                  113702 non-null  str
 26  Genres                      113658 non-null  str
dtypes: bool(3), float64(2), int64(12), str(10)
memory usage: 22.8 MB
```

[494]:
```
dfgames.isnull().sum()
```

[494]:
```
AppID                           0
Name                            1
Release date                    0
Estimated owners                0
Peak CCU                        0
Required age                    0
Price                           0
DiscountDLC count               0
Reviews                         0
Windows                     22263
Mac                             0
Linux                           0
Metacritic score                0
User score                 118355
Positive                        0
Negative                        0
Score rank                      0
Achievements               122571
Recommendations                 0
Average playtime forever   100153
Average playtime two weeks      0
Median playtime forever         0
```

```
Median playtime two weeks          0
Developers                         0
Publishers                      8437
Categories                      8909
Genres                          8953
dtype: int64
```

[495]:
```
dfgames["Windows"].info()
```

```
<class 'pandas.Series'>
RangeIndex: 122611 entries, 0 to 122610
Series name: Windows
Non-Null Count   Dtype
--------------   -----
100348 non-null  str
dtypes: str(1)
memory usage: 958.0 KB
```

[496]:
```
dfgames["Windows"].head(25)
```

[496]:
```
0                              NaN
1           support@mangagamer.com
2               support@8floor.net
3            yujingamesc@gmail.com
4      support@realityexpanded.com
5               office@ignibit.com
6         support@slitherine.co.uk
7     support@summerbearstudio.com
8             heptabeats@gmail.com
9                    info@airem.pl
10                             NaN
11            letodigest@gmail.com
12        support@slitherine.co.uk
13             support@9eyesgs.com
14           nowar1112@hotmail.com
15       support@witchinghour.club
16          ootuka0526@yahoo.co.jp
17     daniel@nightdivestudios.com
18     midnightdash.game@gmail.com
19            catpawgame@gmail.com
20              me@stevennosov.com
21          contact@lunicgames.com
22       megicroomstudio@gmail.com
23     appminded@appmindedapps.com
24                             NaN
Name: Windows, dtype: str
```

[497]:
```
dfgames["Achievements"].sample(25)
```

[497]:
```
1574     NaN
57377    NaN
100469   NaN
93566    NaN
120831   NaN
40673    NaN
117358   NaN
22784    NaN
42388    NaN
101926   NaN
46092    NaN
100666   NaN
29671    NaN
96693    NaN
89175    NaN
29986    NaN
4591     NaN
87151    NaN
90463    NaN
92564    NaN
94815    NaN
108388   NaN
63956    NaN
104979   NaN
88344    NaN
Name: Achievements, dtype: float64
```

[498]:
```
dfgames["Average playtime forever"].sample(25)
```

[498]:
```
19674                                                    NaN
65999                                                    NaN
64713                                                    NaN
101666                                                   NaN
83951                                                    NaN
44319                                                    NaN
62703                                                    NaN
51585                                                    NaN
110599                                                   NaN
12351                                                    NaN
81219                                                    NaN
56098                                                    NaN
39617                                                    NaN
4164                                                     NaN
65636                                                    NaN
81639                 swimsuit, kiss, references to illness.
```

```
109675                                                   NaN
57966                                                    NaN
50529     Mature Content Warning : This Combat-Brawler t…
67942     This game contains mature content, including g…
47621     Enemies And Player Spilling Blood In Gunfights…
68627                          Depictions of blood and gore
97869                                                    NaN
10270                                                    NaN
731                                                      NaN
Name: Average playtime forever, dtype: str
```

[499]:
```
dfgames["User score"].head(50)
```

[499]:
```
0                                                    NaN
1                                                    NaN
2                                                    NaN
3                                                    NaN
4                                                    NaN
5                                                    NaN
6                                                    NaN
7                                                    NaN
8                                                    NaN
9                                                    NaN
10                                                   NaN
11                                                   NaN
12    https://www.metacritic.com/game/pc/fantasy-gen…
13                                                   NaN
14                                                   NaN
15                                                   NaN
16                                                   NaN
17                                                   NaN
18                                                   NaN
19                                                   NaN
20                                                   NaN
21                                                   NaN
22                                                   NaN
23                                                   NaN
24                                                   NaN
25                                                   NaN
26                                                   NaN
27                                                   NaN
28                                                   NaN
29                                                   NaN
30                                                   NaN
31                                                   NaN
32                                                   NaN
33                                                   NaN
```

```
34                                                    NaN
35                                                    NaN
36                                                    NaN
37                                                    NaN
38                                                    NaN
39                                                    NaN
40                                                    NaN
41                                                    NaN
42    https://www.metacritic.com/game/pc/the-metrono…
43                                                    NaN
44                                                    NaN
45                                                    NaN
46                                                    NaN
47                                                    NaN
48                                                    NaN
49                                                    NaN
Name: User score, dtype: str
```

[500]:
```
dfgames["Publishers"].sample(25)
```

[500]:
```
76084                   Nerve Singer
17030                           WATU
22786                      Fred Wood
79853                 Berzerk Studio
8000             [Ad Space Reserved]
59166                Poseidon's kiss
90842                       A Nostru
17264                            NaN
2392                 Artless Gallery
27453     Xiangyu Wu,Xuanwen Chen,V.
4640                           Sicco
112317            Weston Bell-Geddes
117471                     thief7820
35990                 Eldstorm Games
80400                            NaN
36935                 Big Reef Games
89713                   Nihon Falcom
78818                      AVI Games
4731                        MagiKrat
63626                        NaarDev
82344                Jade Ciderquist
121160                     Amotoplay
34402                        SVGames
17389                  Big Robot Ltd
55158                      O.Stogden
Name: Publishers, dtype: str
```

[501]:
```
dfgames["Categories"].sample(25)
```

[501]:
```
47969                          LM Game Studio
92354                                Torcular
17648                                 Banuza
75777                            Game Museum
32256                       Freenzer Network
62167                            Dan Gartman
34400                            MungoBungo1
63032                          ShangkoStudio
6691                        Konnichiwa Games
60682                       Second Dimension
92278                                Tenball
38019                                   TWEN
83641                       ARC SYSTEM WORKS
84202                                Bosonic
76971                                  yocco
70405                   Indie Games Publisher
109624                        GoldenGod Games
931                                Bart Bonte
42200                          Artyom Design
66692                    Brewsterland Studios
98773     THQ Nordic,All Elite Wrestling, LLC
103034                                    NaN
34806                          Labrodex, Inc.
85255
6616                           Undaunted Games
Name: Categories, dtype: str
```

[502]:
```
dfgames["Genres"].sample(25)
```

[502]:
```
10442     Single-player,Full controller support,Family S…
21087                                                  NaN
54627          Single-player,Steam Achievements,Steam Cloud
6718      Single-player,Steam Achievements,Full controll…
14857                         Single-player,Family Sharing
75993                   Multi-player,PvP,Online PvP,LAN PvP
81146     Single-player,Multi-player,Co-op,Shared/Split …
76401                         Single-player,Family Sharing
19213     Single-player,Steam Achievements,Tracked Contr…
68414     Single-player,Steam Achievements,Steam Trading…
261       Multi-player,PvP,Online PvP,Co-op,Online Co-op…
44017     Single-player,Multi-player,PvP,Shared/Split Sc…
8151                          Single-player,Family Sharing
81580       Single-player,Steam Achievements,Family Sharing
48728     Single-player,Steam Achievements,Full controll…
65170     Single-player,Tracked Controller Support,VR On…
```

```
109030    Single-player,Steam Achievements,Captions avai…
21880     Single-player,Steam Achievements,Steam Trading…
98710     Single-player,Steam Achievements,Partial Contr…
108544    Single-player,Multi-player,Cross-Platform Mult…
105254    Single-player,Steam Achievements,Full controll…
55824                          Single-player,Family Sharing
110323    Single-player,Tracked Controller Support,VR On…
67903     Single-player,Steam Achievements,Full controll…
6392      Single-player,Steam Achievements,Steam Trading…
Name: Genres, dtype: str
```

[503]:
```
dfgames = dfgames.drop(columns=[
    'User score',
    'Achievements',
    "Average playtime forever"
])
```

[504]:
```
dfgames.columns
```

[504]:
```
Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
       'Required age', 'Price', 'DiscountDLC count', 'Reviews', 'Windows',
       'Mac', 'Linux', 'Metacritic score', 'Positive', 'Negative',
       'Score rank', 'Recommendations', 'Average playtime two weeks',
       'Median playtime forever', 'Median playtime two weeks', 'Developers',
       'Publishers', 'Categories', 'Genres'],
      dtype='str')
```

[505]:
```
dfgames.shape
```

[505]:
```
(122611, 24)
```

[506]:
```
dfgames["Release date"] = pd.to_datetime(dfgames["Release date"],␣
 ↪errors="coerce")
dfgames = dfgames.drop(columns=["Windows"])
dfgames["Name"] = dfgames["Name"].str.strip().str.lower()
```

[507]:
```
dfgames["Publishers"] = dfgames["Publishers"].fillna("Unknown")
dfgames["Genres"] = dfgames["Genres"].fillna("Unknown")
dfgames["Categories"] = dfgames["Categories"].fillna("Unknown")
dfgames=dfgames.dropna(subset=["Name"])
```

[508]:
```
dfgames2 = dfgames2.reset_index()
```

[509]:
```
dfgames.head()
```

[509]:
```
     AppID                                   Name Release date  \
0  2539430             black dragon mage playtest   2023-08-01
```

```
1   496350  supipara - chapter 1 spring has come!   2016-07-29
2  1034400      mystery solitaire the black raven   2019-05-06
3  3292190                   - vtuber paranoia   2024-10-31
4  3631080                           maze quest vr   2025-04-24

  Estimated owners  Peak CCU  Required age  Price  DiscountDLC count  \
0            0 - 0         0             0   0.00                  0
1        0 - 20000         0             0   5.24                 65
2        0 - 20000         0             0   4.99                  0
3        0 - 20000         1             0   8.99                  0
4        0 - 20000         0             0   4.99                  0

       Reviews   Mac  Linux  Metacritic score  Positive  Negative  Score rank  \
0           []  True  False             False         0         0           0
1           []  True  False             False         0       252           3
2           []  True   True             False         0        21           3
3   ['Korean']  True  False             False         0         0           0
4  ['English']  True  False             False         0         0           0

   Recommendations  Average playtime two weeks  Median playtime forever  \
0                0                           0                        0
1                0                           8                        0
2                0                           0                        0
3               19                           0                        0
4                0                           0                        0

   Median playtime two weeks  Developers            Publishers  \
0                          0           0               Unknown
1                          8           0                minori
2                          0           0           Somer Games
3                          0           0
4                          0           0  Reality Expanded LLC

             Categories                                            Genres
0               Unknown                                           Unknown
1            MangaGamer  Single-player,Steam Trading Cards,Steam Cloud,…
2                Sfloor                       Single-player,Family Sharing
3                       Single-player,Steam Achievements,Family Sharing
4  Reality Expanded LLC  Single-player,VR Only,Steam Leaderboards,Famil…
```

[510]:
```
dfgames.isnull().sum()
```

[510]:
```
AppID                       0
Name                        0
Release date                0
Estimated owners            0
Peak CCU                    0
```

```
Required age                  0
Price                         0
DiscountDLC count             0
Reviews                       0
Mac                           0
Linux                         0
Metacritic score              0
Positive                      0
Negative                      0
Score rank                    0
Recommendations               0
Average playtime two weeks    0
Median playtime forever       0
Median playtime two weeks     0
Developers                    0
Publishers                    0
Categories                    0
Genres                        0
dtype: int64
```

[511]:
```
dfgames.info()
```

```
<class 'pandas.DataFrame'>
Index: 122610 entries, 0 to 122610
Data columns (total 23 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   AppID                       122610 non-null  int64
 1   Name                        122610 non-null  str
 2   Release date                122610 non-null  datetime64[us]
 3   Estimated owners            122610 non-null  str
 4   Peak CCU                    122610 non-null  int64
 5   Required age                122610 non-null  int64
 6   Price                       122610 non-null  float64
 7   DiscountDLC count           122610 non-null  int64
 8   Reviews                     122610 non-null  str
 9   Mac                         122610 non-null  bool
 10  Linux                       122610 non-null  bool
 11  Metacritic score            122610 non-null  bool
 12  Positive                    122610 non-null  int64
 13  Negative                    122610 non-null  int64
 14  Score rank                  122610 non-null  int64
 15  Recommendations             122610 non-null  int64
 16  Average playtime two weeks  122610 non-null  int64
 17  Median playtime forever     122610 non-null  int64
 18  Median playtime two weeks   122610 non-null  int64
 19  Developers                  122610 non-null  int64
```

```
 20  Publishers                  122610 non-null  str
 21  Categories                  122610 non-null  str
 22  Genres                      122610 non-null  str
dtypes: bool(3), datetime64[us](1), float64(1), int64(12), str(6)
memory usage: 20.0 MB
```

[512]:
```
dfgames["Genres"].value_counts().head(10)
```

[512]:
```
Genres
Single-player,Family Sharing
23383
Single-player,Steam Achievements,Family Sharing
9888
Unknown
8952
Single-player
4480
Single-player,Steam Achievements,Steam Cloud,Family Sharing
4202
Single-player,Steam Achievements,Full controller support,Steam Cloud,Family
Sharing     3160
Single-player,Steam Achievements,Full controller support,Family Sharing
3008
Single-player,Full controller support,Family Sharing
2934
Single-player,Partial Controller Support,Family Sharing
2636
Single-player,Tracked Controller Support,VR Only,Family Sharing
1842
Name: count, dtype: int64
```

[513]:
```
dfgames["Categories"].value_counts().head(10)
```

[513]:
```
Categories
Unknown              8908
BFG Entertainment     554
8floor                268
EroticGamesClub       231
Strategy First        206
Conglomerate 5        197
Kagura Games          190
Hede                  184
HH-Games              184
Choice of Games       179
Name: count, dtype: int64
```

[514]:
```
dfgames["Price"].describe()
```

[514]:
```
count    122610.000000
mean          4.765130
std          12.531074
min           0.000000
25%           0.550000
50%           2.240000
75%           5.240000
max         999.980000
Name: Price, dtype: float64
```

[515]:
```
dfgames2.columns
```

[515]:
```
Index(['index', 'Tittle', 'ReleaseDate', 'Rating', 'Genres', 'Platforms',
       'Metascore', 'Userscore', 'Developer', 'Publisher'],
      dtype='str')
```

[516]:
```
dfgames2.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 13913 entries, 0 to 13912
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   index        13913 non-null  int64
 1   Tittle       13913 non-null  str
 2   ReleaseDate  13884 non-null  str
 3   Rating       11610 non-null  str
 4   Genres       13913 non-null  str
 5   Platforms    13912 non-null  str
 6   Metascore    13912 non-null  float64
 7   Userscore    13913 non-null  int64
 8   Developer    13904 non-null  str
 9   Publisher    13911 non-null  str
dtypes: float64(1), int64(2), str(7)
memory usage: 1.1 MB
```

[517]:
```
dfgames2.isnull().sum()
```

[517]:
```
index             0
Tittle            0
ReleaseDate      29
Rating         2303
Genres            0
Platforms         1
Metascore         1
Userscore         0
Developer         9
```

```
Publisher         2
dtype: int64
```

[518]:
```
dfgames2 = dfgames2.rename(columns={"index": "ID"})
```

[519]:
```
dfgames2.isnull().sum()
```

[519]:
```
ID                0
Tittle            0
ReleaseDate      29
Rating         2303
Genres            0
Platforms         1
Metascore         1
Userscore         0
Developer         9
Publisher         2
dtype: int64
```

[520]:
```
dfgames2["Rating"]=dfgames2["Rating"].fillna("Unknown")
dfgames2=dfgames2.
 ↪dropna(subset=["Platforms","Metascore","Developer","Publisher"])
```

[521]:
```
dfgames2.info()
```

```
<class 'pandas.DataFrame'>
Index: 13902 entries, 0 to 13911
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   ID           13902 non-null  int64
 1   Tittle       13902 non-null  str
 2   ReleaseDate  13873 non-null  str
 3   Rating       13902 non-null  str
 4   Genres       13902 non-null  str
 5   Platforms    13902 non-null  str
 6   Metascore    13902 non-null  float64
 7   Userscore    13902 non-null  int64
 8   Developer    13902 non-null  str
 9   Publisher    13902 non-null  str
dtypes: float64(1), int64(2), str(7)
memory usage: 1.2 MB
```

[522]:
```
dfgames2["ReleaseDate"] = pd.to_datetime(dfgames2["ReleaseDate"],␣
 ↪errors="coerce")
dfgames2 = dfgames2.rename(columns={"Tittle": "Title"})
dfgames2["Title"] = dfgames2["Title"].str.strip().str.lower()
```

```
dfgames2["Developer"] = dfgames2["Developer"].str.strip().str.lower()
dfgames2["Publisher"] = dfgames2["Publisher"].str.strip().str.lower()
```

[523]:
```
dfgames2 = dfgames2.drop_duplicates()
```

[524]:
```
dfgames2.info()
```

```
<class 'pandas.DataFrame'>
Index: 13902 entries, 0 to 13911
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   ID           13902 non-null  int64
 1   Title        13902 non-null  str
 2   ReleaseDate  13873 non-null  datetime64[us]
 3   Rating       13902 non-null  str
 4   Genres       13902 non-null  str
 5   Platforms    13902 non-null  str
 6   Metascore    13902 non-null  float64
 7   Userscore    13902 non-null  int64
 8   Developer    13902 non-null  str
 9   Publisher    13902 non-null  str
dtypes: datetime64[us](1), float64(1), int64(2), str(6)
memory usage: 1.2 MB
```

[525]:
```
dfgames2["Metascore"].describe()
```

[525]:
```
count    13902.000000
mean        70.591785
std         12.272827
min         11.000000
25%         64.000000
50%         72.000000
75%         79.000000
max         99.000000
Name: Metascore, dtype: float64
```

[526]:
```
dfgames2["Userscore"].describe()
```

[526]:
```
count    13902.000000
mean        61.887498
std         24.739244
min          0.000000
25%         58.000000
50%         70.000000
75%         78.000000
max        100.000000
```

```
Name: Userscore, dtype: float64
```

[527]:
```python
dfgames2["Genres"].value_counts().head(10)
```

[527]:
```
Genres
Action Adventure      784
2D Platformer         703
Action RPG            688
FPS                   594
JRPG                  389
Real-Time Strategy    381
Open-World Action     368
Survival              352
Adventure             348
Point-and-Click       337
Name: count, dtype: int64
```

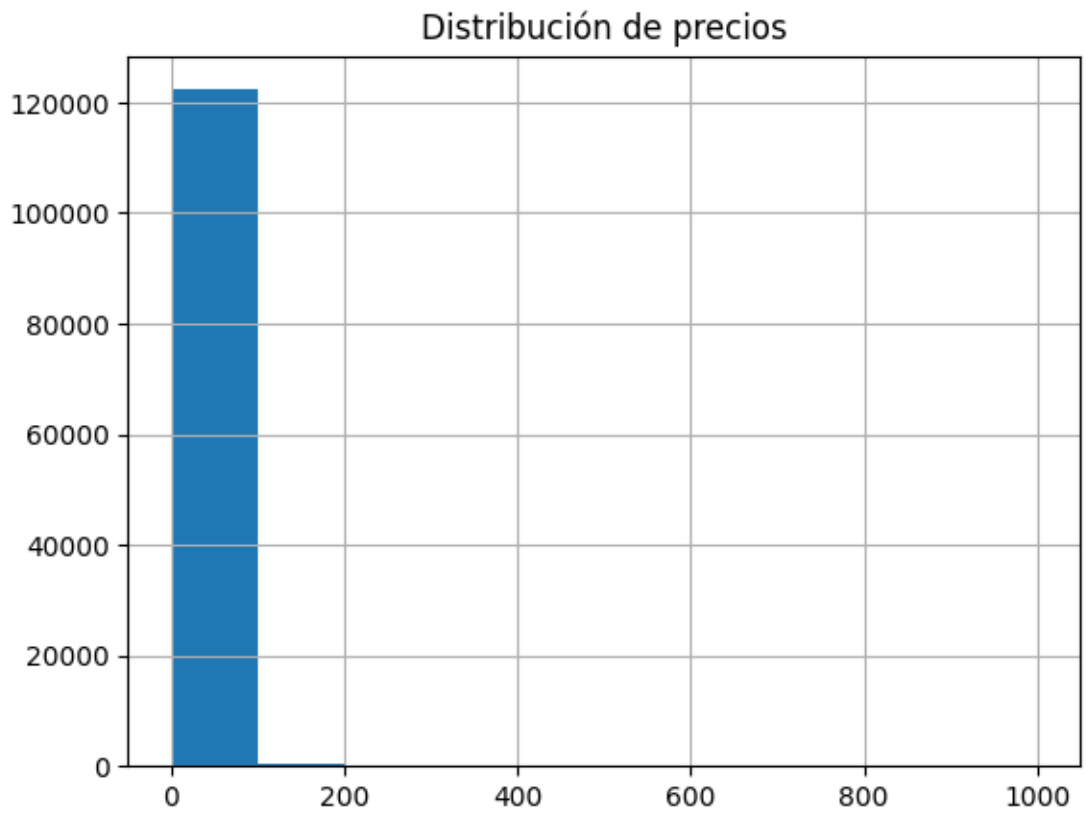
[528]:
```python
dfgames2["Publisher"].value_counts().head(10)
```

[528]:
```
Publisher
nintendo                  479
ubisoft                   364
sega                      334
electronic arts           308
konami                    293
capcom                    255
microsoft game studios    254
square enix               246
activision                208
thq                       156
Name: count, dtype: int64
```

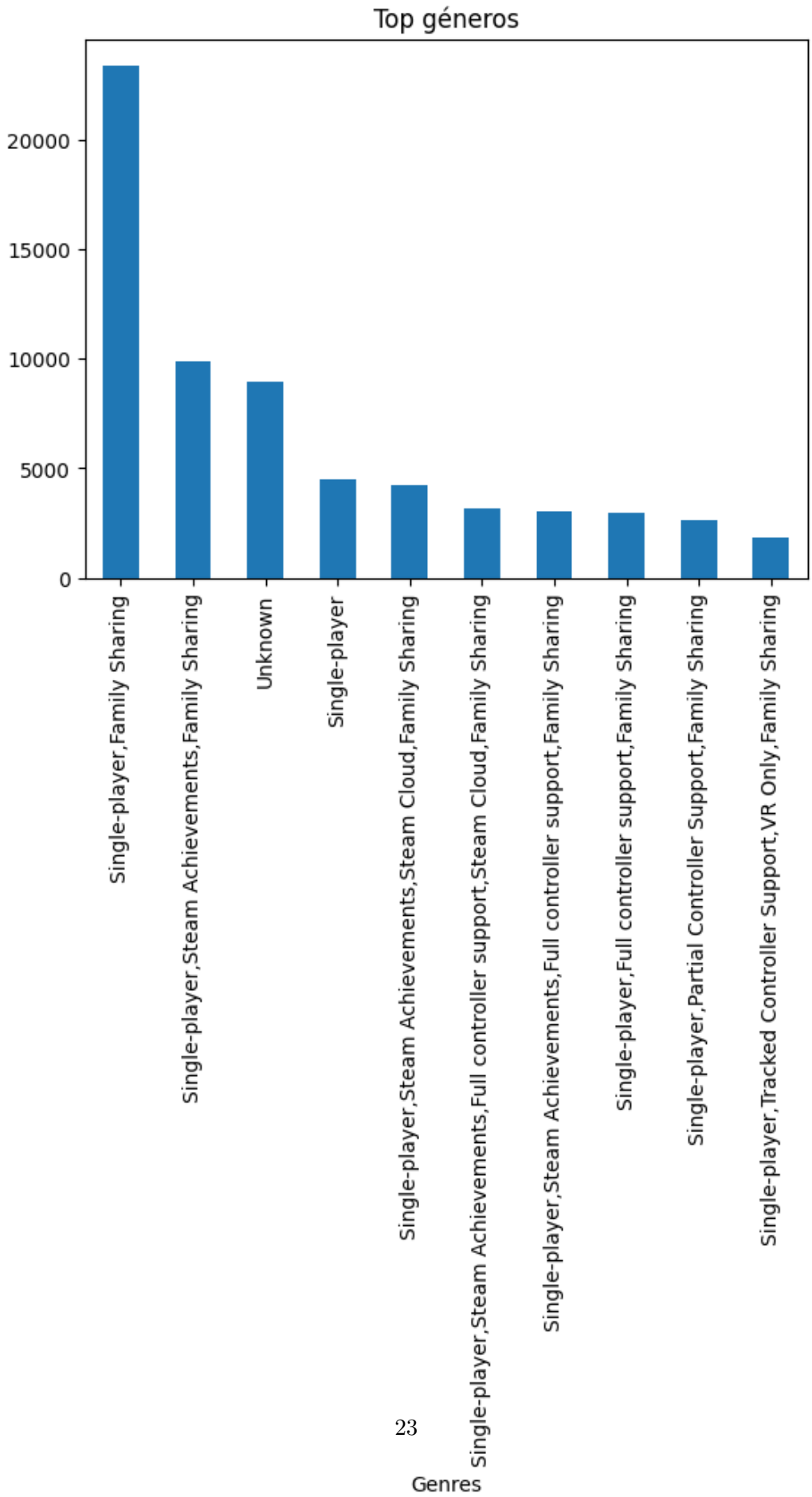
[529]:
```python
dfgames2.groupby("Genres")["Metascore"].mean().sort_values(ascending=False)
```

[529]:
```
Genres
Soccer Management    78.852941
Metroidvania         77.921053
Soccer Sim           77.850746
Baseball Sim         77.409091
Football Sim         76.666667
                        …
Athletics            59.028571
Fishing              57.880000
Sports               57.115385
Hunting              55.000000
Bowling              54.500000
Name: Metascore, Length: 119, dtype: float64
```

[530]:
```
import matplotlib.pyplot as plt

dfgames["Price"].hist()
plt.title("Distribución de precios")
plt.show()
```



[531]:
```
dfgames["Genres"].value_counts().head(10).plot(kind="bar")
plt.title("Top géneros")
plt.show()
```


Top géneros
20000
15000
10000
5000
0
Single-player,Family Sharing
Single-player,Steam Achievements,Family Sharing
Unknown
Single-player
Single-player,Steam Achievements,Steam Cloud,Family Sharing
Single-player,Steam Achievements,Full controller support,Steam Cloud,Family Sharing
Single-player,Steam Achievements,Full controller support,Family Sharing
Single-player,Full controller support,Family Sharing
Single-player,Partial Controller Support,Family Sharing
Single-player,Tracked Controller Support,VR Only,Family Sharing
Genres

[532]:
```
dfgames["AppID"].info()
```

```
<class 'pandas.Series'>
Index: 122610 entries, 0 to 122610
Series name: AppID
Non-Null Count   Dtype
--------------   -----
122610 non-null  int64
dtypes: int64(1)
memory usage: 1.9 MB
```

[533]:
```
dfgames2["ID"].info()
```

```
<class 'pandas.Series'>
Index: 13902 entries, 0 to 13911
Series name: ID
Non-Null Count  Dtype
--------------  -----
13902 non-null  int64
dtypes: int64(1)
memory usage: 217.2 KB
```

[534]:
```
dfgames["rating"] = dfgames["Positive"] / (dfgames["Positive"] +␣
↪dfgames["Negative"])
```

[535]:
```
dfgames["owners_min"] = dfgames["Estimated owners"].str.split("-").str[0].str.
↪replace(",", "").astype(int)
dfgames["owners_max"] = dfgames["Estimated owners"].str.split("-").str[1].str.
↪replace(",", "").astype(int)
```

[536]:
```
dfgames["year"] = dfgames["Release date"].dt.year
```

[537]:
```
dfgames.corr(numeric_only=True)
```

[537]:

	AppID	Peak CCU	Required age	Price	\
AppID	1.000000	-0.014345	-0.096903	-0.007497	
Peak CCU	-0.014345	1.000000	0.030635	0.004664	
Required age	-0.096903	0.030635	1.000000	0.028655	
Price	-0.007497	0.004664	0.028655	1.000000	
DiscountDLC count	-0.192926	0.002021	0.080839	-0.049993	
Mac	0.003120	0.000277	-0.007194	-0.004942	
Linux	-0.159134	0.005591	-0.003083	0.061965	
Metacritic score	-0.104141	0.010781	-0.001882	0.061748	
Positive	-0.018938	-0.000252	0.008382	-0.000949	
Negative	-0.043736	0.846709	0.090763	0.011232	

Score rank	-0.035040	0.830596	0.092754	0.008907
Recommendations	-0.059705	0.004201	0.010484	-0.002821
Average playtime two weeks	-0.015394	0.021905	0.005578	0.003090
Median playtime forever	-0.033750	0.049986	0.048201	0.016441
Median playtime two weeks	-0.011542	0.003201	0.001894	0.002656
Developers	-0.033403	0.027392	0.049131	0.015883
rating	-0.016592	-0.000362	0.002994	-0.002127
owners_min	-0.066669	0.651815	0.113270	0.012126
owners_max	-0.069651	0.635771	0.118740	0.013009
year	0.879927	-0.016259	-0.122649	0.023293

	DiscountDLC count	Mac	Linux	\
AppID	-0.192926	0.003120	-0.159134	
Peak CCU	0.002021	0.000277	0.005591	
Required age	0.080839	-0.007194	-0.003083	
Price	-0.049993	-0.004942	0.061965	
DiscountDLC count	1.000000	0.009364	0.068895	
Mac	0.009364	1.000000	-0.027690	
Linux	0.068895	-0.027690	1.000000	
Metacritic score	0.068131	-0.010777	0.552596	
Positive	0.004226	0.000333	0.003647	
Negative	0.027594	0.000698	0.016282	
Score rank	0.016686	0.000579	0.005763	
Recommendations	0.053209	0.002112	0.013454	
Average playtime two weeks	0.003890	0.000320	0.002261	
Median playtime forever	0.021905	0.000966	0.009196	
Median playtime two weeks	0.002653	0.000254	0.000818	
Developers	0.021231	0.000947	0.008812	
rating	-0.000259	0.000262	0.002738	
owners_min	0.035126	0.001036	0.021547	
owners_max	0.037581	0.001076	0.022699	
year	-0.170111	0.005843	-0.164042	

	Metacritic score	Positive	Negative	Score rank	\
AppID	-0.104141	-0.018938	-0.043736	-0.035040	
Peak CCU	0.010781	-0.000252	0.846709	0.830596	
Required age	-0.001882	0.008382	0.090763	0.092754	
Price	0.061748	-0.000949	0.011232	0.008907	
DiscountDLC count	0.068131	0.004226	0.027594	0.016686	
Mac	-0.010777	0.000333	0.000698	0.000579	
Linux	0.552596	0.003647	0.016282	0.005763	
Metacritic score	1.000000	0.005956	0.021139	0.009338	
Positive	0.005956	1.000000	-0.000592	-0.000443	
Negative	0.021139	-0.000592	1.000000	0.818396	
Score rank	0.009338	-0.000443	0.818396	1.000000	
Recommendations	0.015970	0.001651	0.013942	0.010422	
Average playtime two weeks	0.000302	-0.000045	0.021641	0.021422	

Median playtime forever	0.004665	-0.000747	0.061724	0.058395
Median playtime two weeks	-0.000153	-0.000026	0.004258	0.003983
Developers	0.005385	-0.000623	0.036839	0.036416
rating	0.004640	0.916497	-0.000920	-0.000752
owners_min	0.023902	-0.000868	0.734410	0.778662
owners_max	0.024301	-0.000868	0.734566	0.768734
year	-0.107036	-0.019918	-0.049555	-0.035738

	Recommendations	Average playtime two weeks \
AppID	-0.059705	-0.015394
Peak CCU	0.004201	0.021905
Required age	0.010484	0.005578
Price	-0.002821	0.003090
DiscountDLC count	0.053209	0.003890
Mac	0.002112	0.000320
Linux	0.013454	0.002261
Metacritic score	0.015970	0.000302
Positive	0.001651	-0.000045
Negative	0.013942	0.021641
Score rank	0.010422	0.021422
Recommendations	1.000000	0.002848
Average playtime two weeks	0.002848	1.000000
Median playtime forever	0.011362	0.335276
Median playtime two weeks	0.001522	0.994210
Developers	0.010214	0.309575
rating	0.000718	-0.000275
owners_min	0.018618	0.024768
owners_max	0.019278	0.024925
year	-0.051538	-0.015161

	Median playtime forever \
AppID	-0.033750
Peak CCU	0.049986
Required age	0.048201
Price	0.016441
DiscountDLC count	0.021905
Mac	0.000966
Linux	0.009196
Metacritic score	0.004665
Positive	-0.000747
Negative	0.061724
Score rank	0.058395
Recommendations	0.011362
Average playtime two weeks	0.335276
Median playtime forever	1.000000
Median playtime two weeks	0.303078
Developers	0.973052

rating	-0.001143
owners_min	0.096996
owners_max	0.098853
year	-0.028221

	Median playtime two weeks	Developers	rating	\
AppID	-0.011542	-0.033403	-0.016592	
Peak CCU	0.003201	0.027392	-0.000362	
Required age	0.001894	0.049131	0.002994	
Price	0.002656	0.015883	-0.002127	
DiscountDLC count	0.002653	0.021231	-0.000259	
Mac	0.000254	0.000947	0.000262	
Linux	0.000818	0.008812	0.002738	
Metacritic score	-0.000153	0.005385	0.004640	
Positive	-0.000026	-0.000623	0.916497	
Negative	0.004258	0.036839	-0.000920	
Score rank	0.003983	0.036416	-0.000752	
Recommendations	0.001522	0.010214	0.000718	
Average playtime two weeks	0.994210	0.309575	-0.000275	
Median playtime forever	0.303078	0.973052	-0.001143	
Median playtime two weeks	1.000000	0.278736	-0.000199	
Developers	0.278736	1.000000	-0.001070	
rating	-0.000199	-0.001070	1.000000	
owners_min	0.004722	0.064867	-0.001458	
owners_max	0.004783	0.065760	-0.001476	
year	-0.010921	-0.028338	-0.016494	

	owners_min	owners_max	year
AppID	-0.066669	-0.069651	0.879927
Peak CCU	0.651815	0.635771	-0.016259
Required age	0.113270	0.118740	-0.122649
Price	0.012126	0.013009	0.023293
DiscountDLC count	0.035126	0.037581	-0.170111
Mac	0.001036	0.001076	0.005843
Linux	0.021547	0.022699	-0.164042
Metacritic score	0.023902	0.024301	-0.107036
Positive	-0.000868	-0.000868	-0.019918
Negative	0.734410	0.734566	-0.049555
Score rank	0.778662	0.768734	-0.035738
Recommendations	0.018618	0.019278	-0.051538
Average playtime two weeks	0.024768	0.024925	-0.015161
Median playtime forever	0.096996	0.098853	-0.028221
Median playtime two weeks	0.004722	0.004783	-0.010921
Developers	0.064867	0.065760	-0.028338
rating	-0.001458	-0.001476	-0.016494
owners_min	1.000000	0.996050	-0.078050
owners_max	0.996050	1.000000	-0.080854

```
year                          -0.078050   -0.080854  1.000000
```

[538]:
```
dfgames.groupby(pd.cut(dfgames["Price"], bins=5))["Recommendations"].mean()
```

[538]:
```
Price
(-1.0, 199.996]       18.088195
(199.996, 399.992]     0.000000
(399.992, 599.988]     0.000000
(799.984, 999.98]      0.000000
Name: Recommendations, dtype: float64
```

[539]:
```
dfgames.sort_values("Peak CCU", ascending=False).head(10)
```

[539]:
```
          AppID                   Name Release date       Estimated owners  \
45509       730       counter-strike 2   2012-08-21  100000000 - 200000000
4193        570                  dota 2   2013-07-09  100000000 - 200000000
8878     578080    pubg: battlegrounds   2017-12-21  100000000 - 200000000
104347  3419430              bongo cat   2025-03-05              0 - 20000
17581   2622380  elden ring nightreign   2025-05-29      2000000 - 5000000
5377     252490                   rust   2018-02-08    20000000 - 50000000
54502   1172470          apex legends   2020-11-04  100000000 - 200000000
109015  2767030          marvel rivals   2024-12-05              0 - 20000
86168    431960       wallpaper engine   2018-11-16    20000000 - 50000000
15313   1671210              deltarune   2025-06-04      1000000 - 2000000

        Peak CCU  Required age  Price  DiscountDLC count  \
45509    1013936             0   0.00                  0
4193      623941             0   0.00                  0
8878      314682            13   0.00                  0
104347    164168             0   0.00                  0
17581     163599            17  29.99                 25
5377      143870             0  19.99                 50
54502     124262             0   0.00                  0
109015    106611             0   0.00                  0
86168      91184             0   3.99                 20
15313      83936             0  19.99                 20

                                                Reviews   Mac  Linux  \
45509                          ['English', 'Indonesian']  True  False
4193    ['English', 'Korean', 'Simplified Chinese', 'V…  True   True
8878                                                  []  True  False
104347                                                []  True  False
17581                 ['English', 'Traditional Chinese']  True  False
5377    ['English', 'French', 'Italian', 'German', 'Sp…  True   True
54502   ['English', 'French', 'Italian', 'German', 'Sp…  True  False
109015  ['English', 'Simplified Chinese', 'Japanese', …  True  False
86168                                                 []  True  False
```

15313				[]	True	True

	Metacritic score	Positive	Negative	Score rank	Recommendations	\
45509	True	0	7642084	1173003	1	
4193	True	0	2037143	461826	0	
8878	False	0	1520457	1037487	37	
104347	False	0	0	0	7	
17581	False	0	48901	16771	37	
5377	False	0	1071135	156649	92	
54502	False	0	668053	326926	12	
109015	False	0	228817	58752	49	
86168	False	0	876898	17560	17	
15313	False	0	2804	31	0	

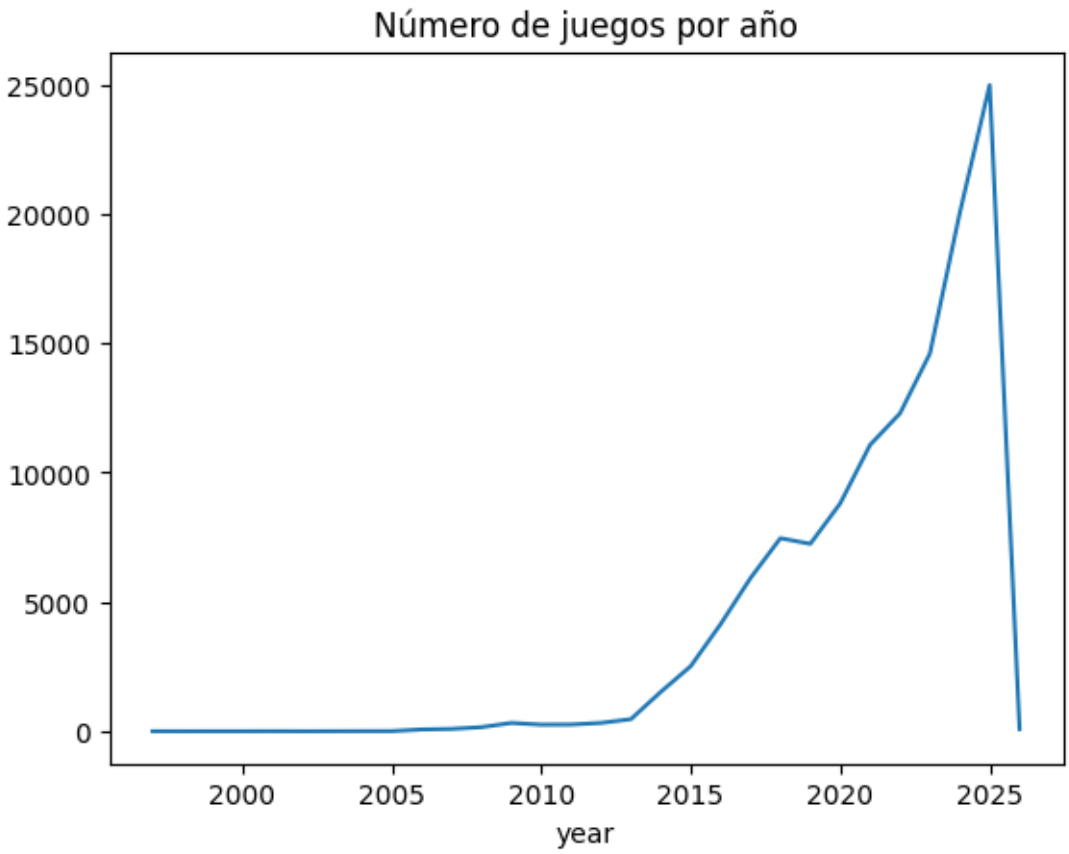
	Average playtime two weeks	Median playtime forever	\
45509	33906	702	
4193	54064	1427	
8878	23787	730	
104347	33409	7434	
17581	4138	857	
5377	24264	1246	
54502	10073	615	
109015	2853	554	
86168	4378	135	
15313	1283	109	

	Median playtime two weeks	Developers	Publishers	\
45509	6237	307	Valve	
4193	1155	860	Valve	
8878	6082	302	PUBG Corporation	
104347	2225	4891	Marcel Zurawka,Julius Krüger	
17581	2242	432	FromSoftware, Inc.	
5377	3044	373	Facepunch Studios	
54502	916	269	Respawn	
109015	498	233	NetEase Games	
86168	1045	9	Wallpaper Engine Team	
15313	1160	151	tobyfox	

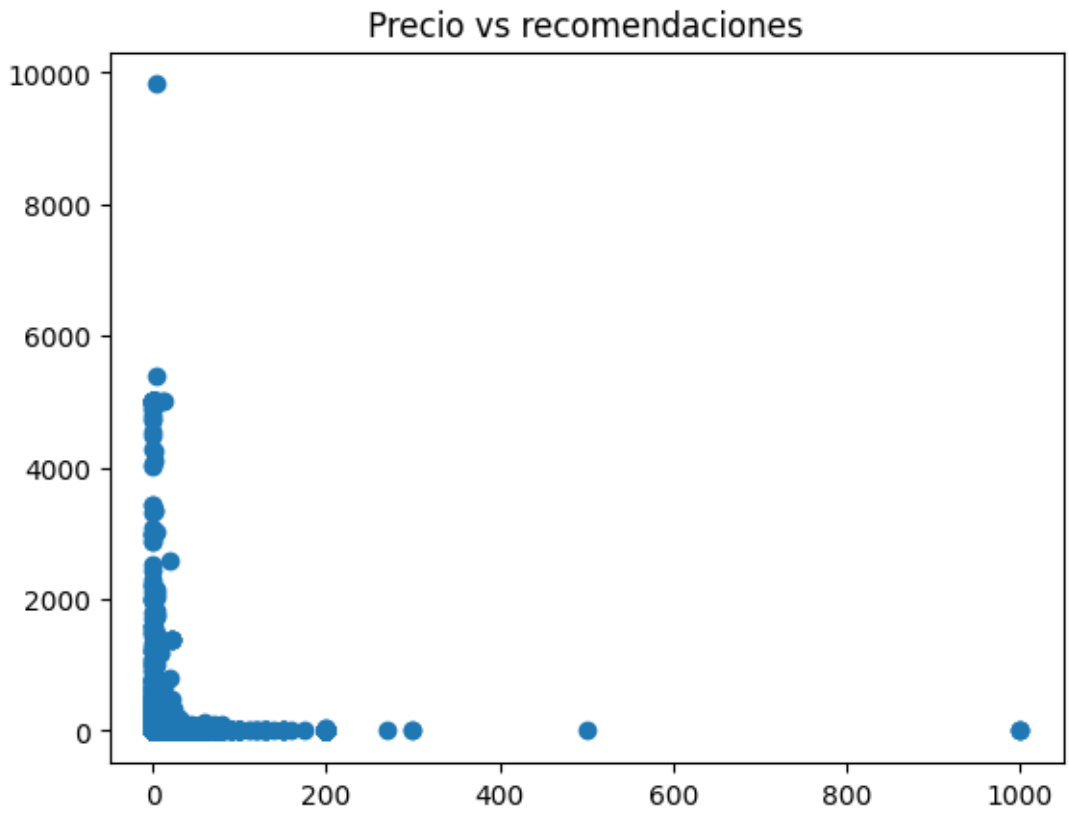
	Categories	\
45509	Valve	
4193	Valve	
8878	KRAFTON, Inc.	
104347	Irox Games	
17581	FromSoftware, Inc.,Bandai Namco Entertainment	
5377	Facepunch Studios	
54502	Electronic Arts	
109015	NetEase Games	

```
86168                          Wallpaper Engine Team
15313                                       tobyfox

                                                Genres  rating  owners_min  \
45509   Multi-player,Cross-Platform Multiplayer,Steam …     0.0   100000000
4193    Multi-player,Co-op,Steam Trading Cards,Steam W…     0.0   100000000
8878    Multi-player,PvP,Online PvP,Stats,Remote Play …     0.0   100000000
104347  Single-player,Multi-player,Co-op,Online Co-op,…     NaN           0
17581   Single-player,Multi-player,Co-op,Online Co-op,…     0.0     2000000
5377    Multi-player,MMO,PvP,Online PvP,Co-op,Online C…     0.0    20000000
54502   Multi-player,PvP,Online PvP,Co-op,Online Co-op…     0.0   100000000
109015  Multi-player,PvP,Online PvP,Co-op,Online Co-op…     0.0           0
86168   Steam Achievements,Steam Trading Cards,Steam W…     0.0    20000000
15313   Single-player,Full controller support,Steam Cl…     0.0     1000000

        owners_max  year
45509    200000000  2012
4193     200000000  2013
8878     200000000  2017
104347       20000  2025
17581      5000000  2025
5377      50000000  2018
54502    200000000  2020
109015       20000  2024
86168     50000000  2018
15313      2000000  2025
```

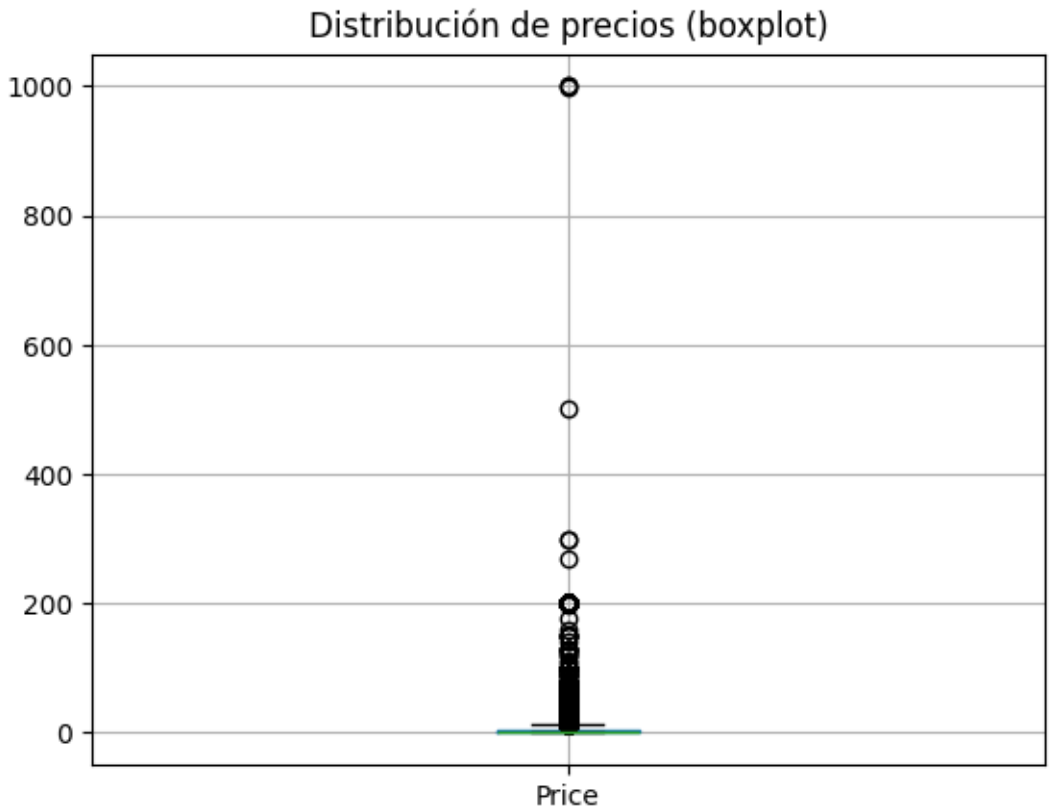
[540]:
```
dfgames["year"].value_counts().sort_index().plot()
plt.title("Número de juegos por año")
plt.show()
```



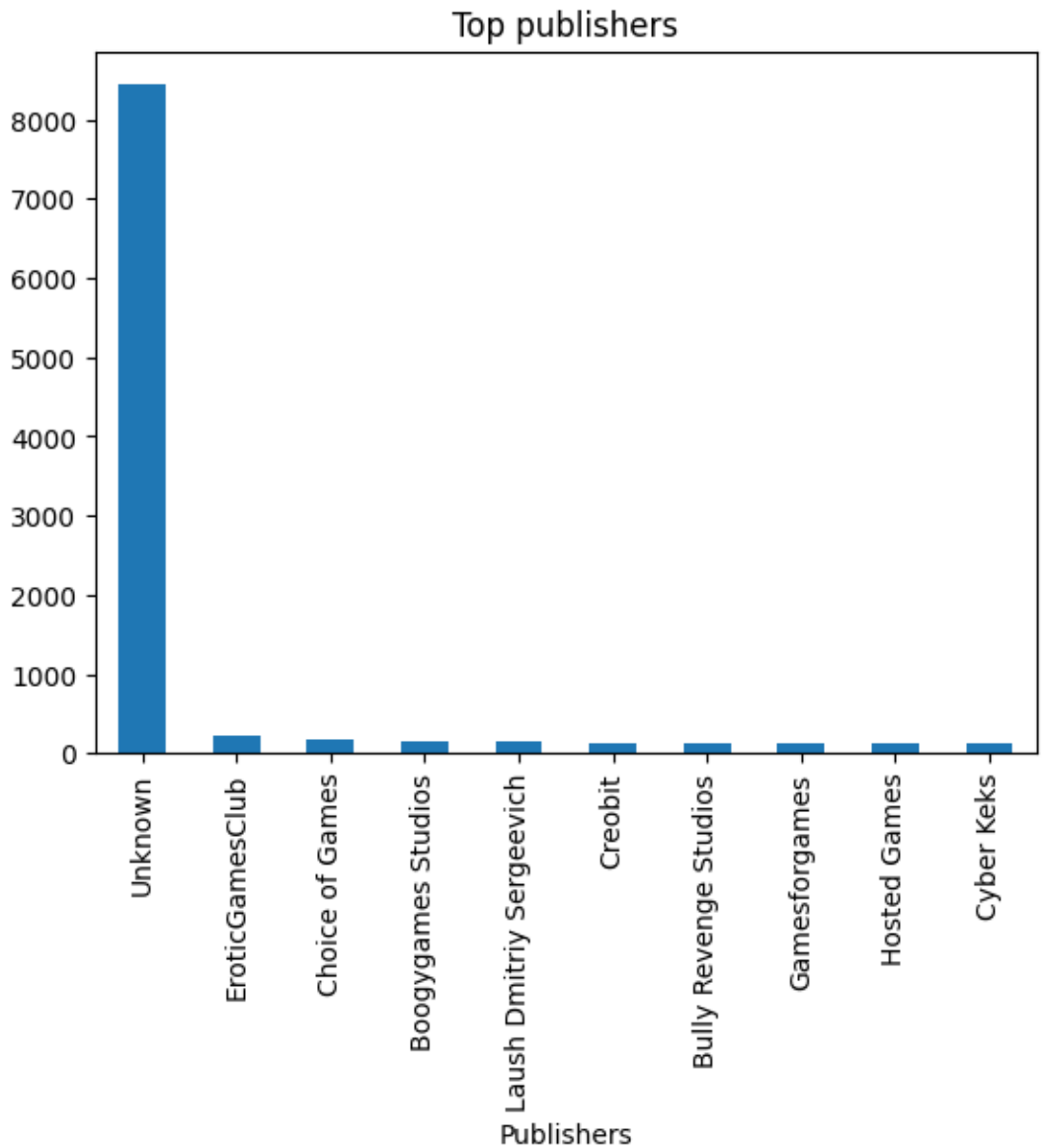
[541]:
```
plt.scatter(dfgames["Price"], dfgames["Recommendations"])
plt.title("Precio vs recomendaciones")
plt.show()
```



[542]:
```
dfgames.boxplot(column="Price")
plt.title("Distribución de precios (boxplot)")
plt.show()
```



[543]:
```
dfgames["Publishers"].value_counts().head(10).plot(kind="bar")
plt.title("Top publishers")
plt.show()
```



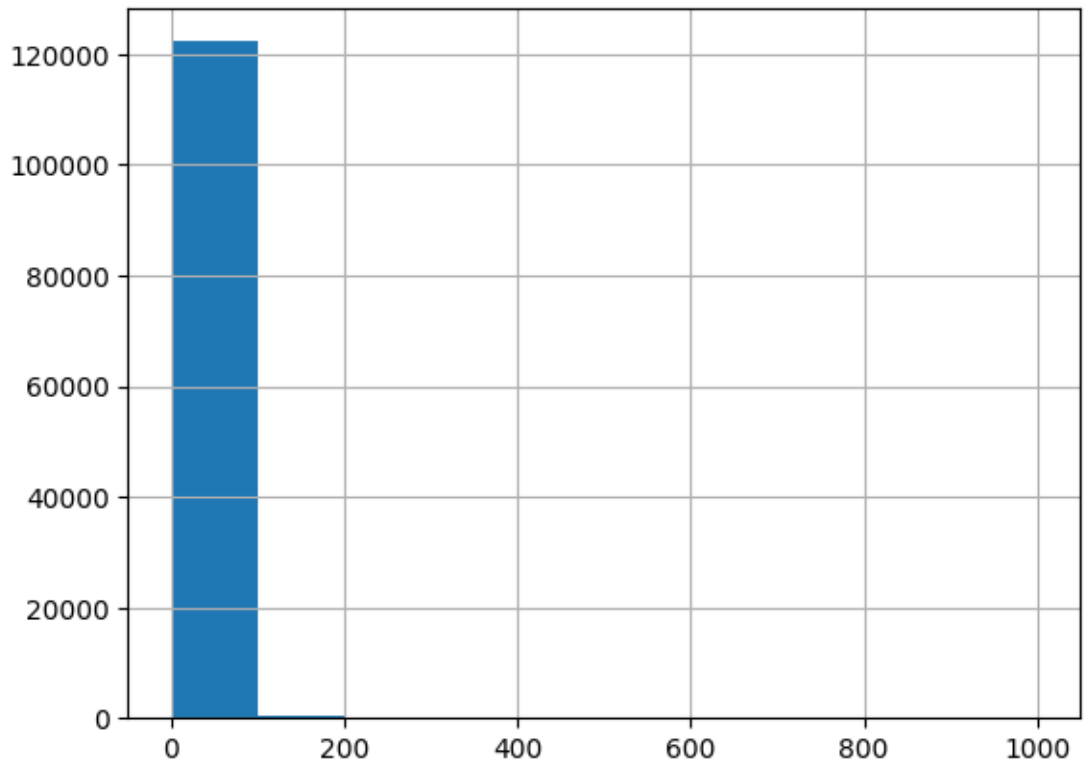
[544]:
```
dfgames["Name_clean"] = dfgames["Name"].str.lower().str.strip()
dfgames2["Title_clean"] = dfgames2["Title"].str.lower().str.strip()
```

[545]:
```
df_final = dfgames.merge(
    dfgames2,
    left_on="Name_clean",
    right_on="Title_clean",
    how="left"
)
```

[546]:
```
df_final = df_final.drop(columns=["Name_clean", "Title_clean"])
```

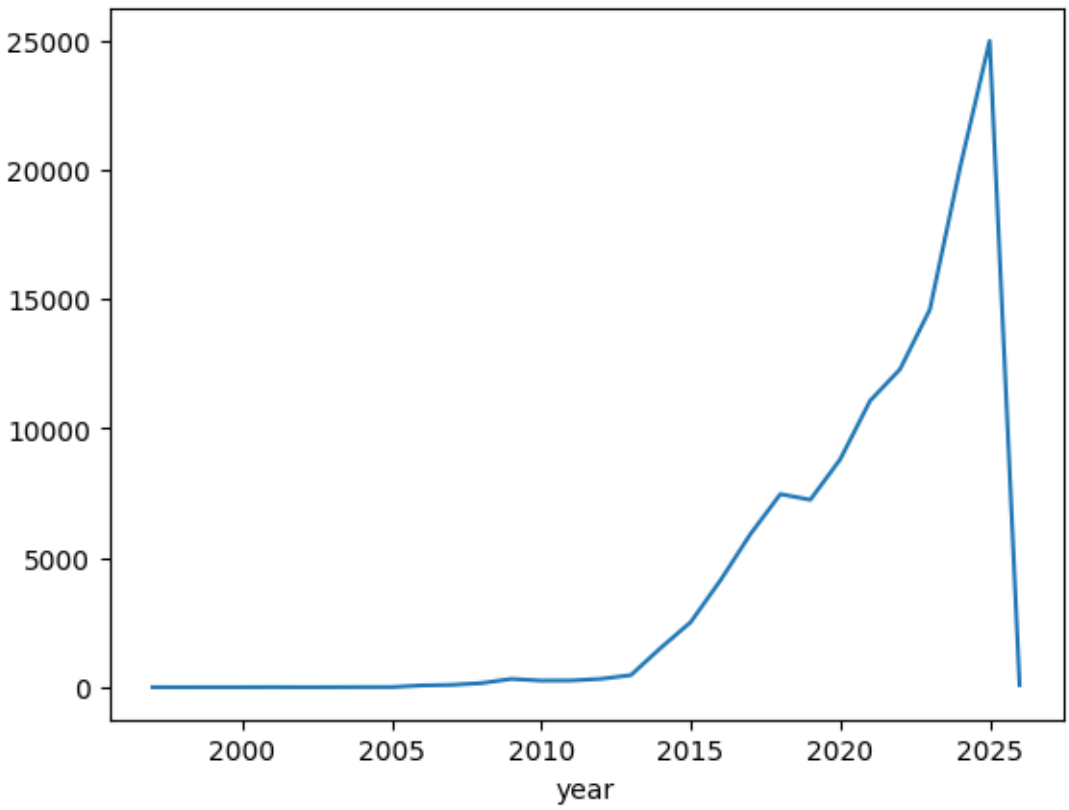
[547]:
```
df_final.describe()
df_final["Price"].hist()
```

[547]: <Axes: >



[548]:
```
df_final["year"].value_counts().sort_index().plot()
```

[548]: <Axes: xlabel='year'>



[549]:
```
df_final.sort_values("Peak CCU", ascending=False).head(10)
```

[549]:
```
          AppID                   Name Release date       Estimated owners  \
45508       730       counter-strike 2   2012-08-21  100000000 - 200000000
4193        570                  dota 2   2013-07-09  100000000 - 200000000
8878     578080    pubg: battlegrounds   2017-12-21  100000000 - 200000000
104346  3419430               bongo cat   2025-03-05              0 - 20000
17581   2622380  elden ring nightreign   2025-05-29      2000000 - 5000000
5377     252490                    rust   2018-02-08    20000000 - 50000000
54501   1172470           apex legends   2020-11-04  100000000 - 200000000
109014  2767030           marvel rivals   2024-12-05              0 - 20000
86167    431960       wallpaper engine   2018-11-16    20000000 - 50000000
15313   1671210               deltarune   2025-06-04      1000000 - 2000000

        Peak CCU  Required age  Price  DiscountDLC count  \
45508    1013936             0   0.00                  0
4193      623941             0   0.00                  0
8878      314682            13   0.00                  0
104346    164168             0   0.00                  0
17581     163599            17  29.99                 25
5377      143870             0  19.99                 50
```

```
54501     124262             0   0.00                  0
109014    106611             0   0.00                  0
86167      91184             0   3.99                 20
15313      83936             0  19.99                 20

                                                Reviews   Mac  Linux  \
45508                            ['English', 'Indonesian']  True  False
4193    ['English', 'Korean', 'Simplified Chinese', 'V…  True   True
8878                                                   []  True  False
104346                                                 []  True  False
17581                   ['English', 'Traditional Chinese']  True  False
5377    ['English', 'French', 'Italian', 'German', 'Sp…  True   True
54501   ['English', 'French', 'Italian', 'German', 'Sp…  True  False
109014  ['English', 'Simplified Chinese', 'Japanese', …  True  False
86167                                                  []  True  False
15313                                                  []  True   True

        Metacritic score  Positive  Negative  Score rank  Recommendations  \
45508               True         0   7642084     1173003                1
4193                True         0   2037143      461826                0
8878               False         0   1520457     1037487               37
104346             False         0         0           0                7
17581              False         0     48901       16771               37
5377               False         0   1071135      156649               92
54501              False         0    668053      326926               12
109014             False         0    228817       58752               49
86167              False         0    876898       17560               17
15313              False         0      2804          31                0

        Average playtime two weeks  Median playtime forever  \
45508                        33906                      702
4193                         54064                     1427
8878                         23787                      730
104346                       33409                     7434
17581                         4138                      857
5377                         24264                     1246
54501                        10073                      615
109014                        2853                      554
86167                         4378                      135
15313                         1283                      109

        Median playtime two weeks  Developers                     Publishers  \
45508                        6237         307                          Valve
4193                         1155         860                          Valve
8878                         6082         302                PUBG Corporation
104346                       2225        4891  Marcel Zurawka,Julius Krüger
17581                        2242         432             FromSoftware, Inc.
```

```
5377                        3044         373             Facepunch Studios
54501                        916         269                       Respawn
109014                       498         233                  NetEase Games
86167                       1045           9          Wallpaper Engine Team
15313                       1160         151                       tobyfox

                                       Categories  \
45508                                       Valve
4193                                        Valve
8878                                 KRAFTON, Inc.
104346                                  Irox Games
17581   FromSoftware, Inc.,Bandai Namco Entertainment
5377                              Facepunch Studios
54501                               Electronic Arts
109014                                NetEase Games
86167                           Wallpaper Engine Team
15313                                        tobyfox

                                            Genres_x  rating  owners_min  \
45508   Multi-player,Cross-Platform Multiplayer,Steam …     0.0   100000000
4193    Multi-player,Co-op,Steam Trading Cards,Steam W…     0.0   100000000
8878    Multi-player,PvP,Online PvP,Stats,Remote Play …     0.0   100000000
104346  Single-player,Multi-player,Co-op,Online Co-op,…     NaN           0
17581   Single-player,Multi-player,Co-op,Online Co-op,…     0.0     2000000
5377    Multi-player,MMO,PvP,Online PvP,Co-op,Online C…     0.0    20000000
54501   Multi-player,PvP,Online PvP,Co-op,Online Co-op…     0.0   100000000
109014  Multi-player,PvP,Online PvP,Co-op,Online Co-op…     0.0           0
86167   Steam Achievements,Steam Trading Cards,Steam W…     0.0    20000000
15313   Single-player,Full controller support,Steam Cl…     0.0     1000000

        owners_max  year      ID                  Title ReleaseDate   Rating  \
45508    200000000  2012  2361.0       counter-strike 2  2023-09-27  Unknown
4193     200000000  2013   354.0                 dota 2  2013-07-09  Unknown
8878     200000000  2017     NaN                    NaN         NaT      NaN
104346       20000  2025     NaN                    NaN         NaT      NaN
17581      5000000  2025  4413.0  elden ring nightreign  2025-05-30        M
5377      50000000  2018  8562.0                   rust  2018-02-08  Unknown
54501    200000000  2020     NaN                    NaN         NaT      NaN
109014       20000  2024  6378.0          marvel rivals  2024-12-06  Unknown
86167     50000000  2018     NaN                    NaN         NaT      NaN
15313      2000000  2025  1057.0              deltarune  2025-06-05        T

                            Genres_y                       Platforms  \
45508                    Tactical FPS                              PC
4193                             MOBA                              PC
8878                              NaN                             NaN
104346                            NaN                             NaN
```

```
17581                      Action RPG  PlayStation 5,Xbox Series X,PC
5377                         Survival       PC,Xbox One,PlayStation 4
54501                             NaN                              NaN
109014  Tactical Third Person Shooter  PC,PlayStation 5,Xbox Series X
86167                             NaN                              NaN
15313                            JRPG               Nintendo Switch 2

        Metascore  Userscore          Developer  \
45508        82.0       53.0     valve software
4193         90.0       65.0     valve software
8878          NaN        NaN                NaN
104346        NaN        NaN                NaN
17581        77.0       67.0      from software
5377         69.0       65.0  facepunch studios
54501         NaN        NaN                NaN
109014       74.0       69.0      netease games
86167         NaN        NaN                NaN
15313        86.0       90.0                8-4

                              Publisher
45508                     valve software
4193                valve software,nexon
8878                                 NaN
104346                               NaN
17581   bandai namco games,from software
5377                   facepunch studios
54501                                NaN
109014                     netease games
86167                                NaN
15313                                8-4
```

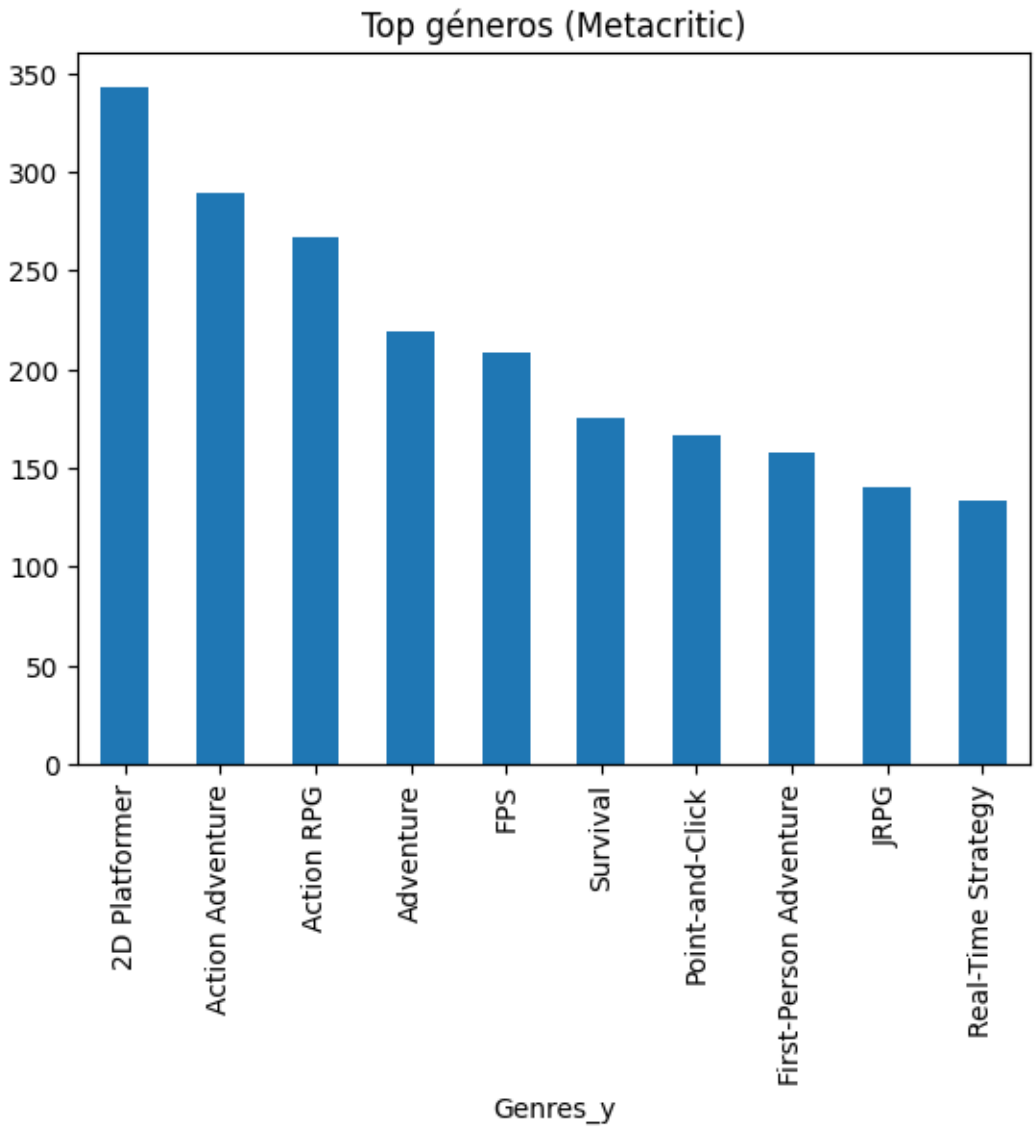
[550]:
```
df_final.groupby("Genres_y")["Metascore"].mean().sort_values(ascending=False).
↪head(10)
```

[550]:
```
Genres_y
Exercise             89.000000
Baseball             83.000000
Basketball Sim       82.666667
Card Battle          79.790698
Text Adventure       77.800000
Logic Puzzle         77.750000
Arcade Racing        77.714286
Metroidvania         77.618182
Compilation          77.533333
Soccer Management    77.500000
Name: Metascore, dtype: float64
```

[551]:
```
df_final.groupby("Genres_y")["Userscore"].mean().sort_values(ascending=False).
↪head(10)
```

[551]:
```
Genres_y
Racing                81.000000
Trivia                78.000000
Racing Sim            76.000000
Dancing               74.000000
Artillery             73.400000
Stacking Puzzle       73.200000
Train Sim             72.500000
Real-Time Tactics     72.400000
MOBA                  72.200000
Western RPG           71.216216
Name: Userscore, dtype: float64
```

[552]:
```
df_final["Genres_y"].value_counts().head(10).plot(kind="bar")
plt.title("Top géneros (Metacritic)")
plt.show()
```



[553]:
```
df_final["rating"].describe()
```

[553]:
```
count    80339.000000
mean         0.000303
std          0.014677
min          0.000000
25%          0.000000
50%          0.000000
75%          0.000000
max          0.909091
Name: rating, dtype: float64
```

[554]:
```python
df_final.sort_values("rating", ascending=False).head(10)
```

[554]:

	AppID	Name	Release date
48514	471720	k station	2016-06-15
65133	891870	king of phoenix	2018-09-27
93789	929290	strip breaker : hentai girls	2018-09-30
27656	961640	the tower of five hearts	2018-11-14
118362	929310	kamasutra connect : sexy hentai girls	2018-11-13
38005	946800	hentai 2+2=4	2018-10-05
51397	1013180	funbag fantasy	2019-01-31
71760	966460	undress tournament	2019-01-24
22730	677730	karmasutra	2017-09-29
83973	1014730	cyndy	2019-05-16

	Estimated owners	Peak CCU	Required age	Price	DiscountDLC count
48514	0 - 20000	0	0	0.00	0
65133	0 - 20000	0	0	2.49	50
93789	0 - 20000	0	0	0.49	51
27656	0 - 20000	0	0	7.49	0
118362	0 - 20000	0	0	0.49	51
38005	0 - 20000	0	0	0.49	0
51397	0 - 20000	1	0	29.24	35
71760	0 - 20000	0	0	0.00	0
22730	0 - 20000	0	0	8.99	40
83973	0 - 20000	0	0	7.99	0

	Reviews	Mac	Linux	Metacritic score	Positive	Negative
48514	[]	True	False	False	100	10
65133	[]	True	True	False	84	11
93789	[]	True	False	False	53	8
27656	[]	True	True	True	100	16
118362	[]	True	False	False	68	11
38005	[]	True	False	False	61	10
51397	[]	True	False	False	100	21
71760	[]	True	False	False	57	12
22730	[]	True	True	True	60	13
83973	['English']	True	False	False	68	16

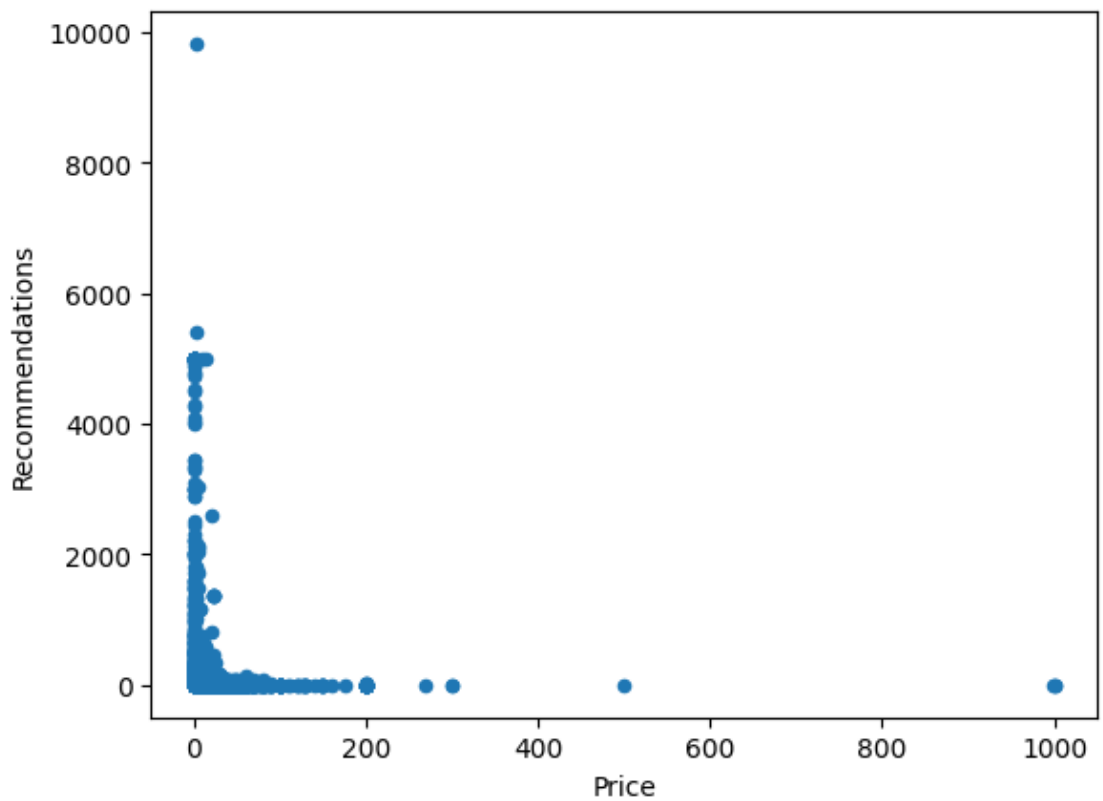
	Score rank	Recommendations	Average playtime two weeks
48514	0	0	0
65133	3	10	38
93789	7	0	0
27656	0	0	372
118362	5	18	0
38005	6	16	0
51397	0	0	282
71760	9	10	0

```
22730            6                36                          17
83973            6                21                           5

        Median playtime forever  Median playtime two weeks  Developers  \
48514                         0                          0           0
65133                         0                         38           0
93789                         0                          0           0
27656                         0                        372           0
118362                        0                          0           0
38005                         0                          0           0
51397                         0                        282           0
71760                         0                          0           0
22730                         0                         17           0
83973                         0                          5           0

                             Publishers         Categories  \
48514                        Maya Games         Maya Games
65133                     King Key Games     King Key Games
93789                          Playtouch          Playtouch
27656                          TsukiWare         MangaGamer
118362                         Playtouch          Playtouch
38005                       AFBIK Studio       AFBIK Studio
51397                             Waffle         MangaGamer
71760   HawkX Games,Double Mirage Studio        HawkX Games
22730                     Top Hat Curios  Nutaku Publishing
83973                     DreamBig Games     DreamBig Games

                                                Genres_x    rating  \
48514        Single-player,Captions available,Steam Cloud  0.909091
65133   Single-player,Steam Achievements,Steam Trading…  0.884211
93789                       Single-player,Family Sharing  0.868852
27656                       Single-player,Family Sharing  0.862069
118362    Single-player,Steam Achievements,Family Sharing  0.860759
38005   Single-player,Steam Achievements,Steam Cloud,F…  0.859155
51397            Single-player,Steam Cloud,Family Sharing  0.826446
71760   Single-player,Steam Achievements,In-App Purcha…  0.826087
22730   Single-player,Steam Achievements,Steam Trading…  0.821918
83973     Single-player,Steam Achievements,Family Sharing  0.809524

        owners_min  owners_max  year  ID Title ReleaseDate Rating Genres_y  \
48514            0       20000  2016 NaN   NaN         NaT    NaN      NaN
65133            0       20000  2018 NaN   NaN         NaT    NaN      NaN
93789            0       20000  2018 NaN   NaN         NaT    NaN      NaN
27656            0       20000  2018 NaN   NaN         NaT    NaN      NaN
118362           0       20000  2018 NaN   NaN         NaT    NaN      NaN
38005            0       20000  2018 NaN   NaN         NaT    NaN      NaN
51397            0       20000  2019 NaN   NaN         NaT    NaN      NaN
```

```
71760            0       20000  2019 NaN   NaN         NaT    NaN      NaN
22730            0       20000  2017 NaN   NaN         NaT    NaN      NaN
83973            0       20000  2019 NaN   NaN         NaT    NaN      NaN

       Platforms  Metascore  Userscore Developer Publisher
48514        NaN        NaN        NaN       NaN       NaN
65133        NaN        NaN        NaN       NaN       NaN
93789        NaN        NaN        NaN       NaN       NaN
27656        NaN        NaN        NaN       NaN       NaN
118362       NaN        NaN        NaN       NaN       NaN
38005        NaN        NaN        NaN       NaN       NaN
51397        NaN        NaN        NaN       NaN       NaN
71760        NaN        NaN        NaN       NaN       NaN
22730        NaN        NaN        NaN       NaN       NaN
83973        NaN        NaN        NaN       NaN       NaN
```

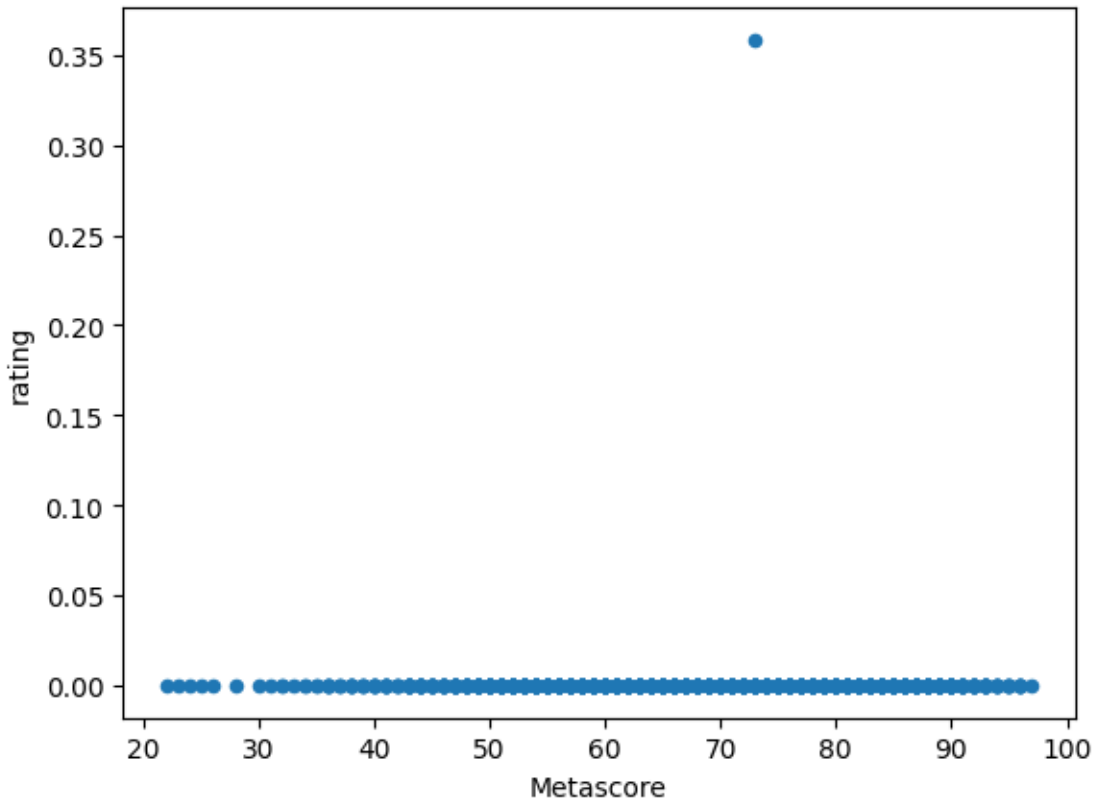
[555]:
```
df_final.plot.scatter(x="Price", y="Recommendations")
```

[555]: <Axes: xlabel='Price', ylabel='Recommendations'>



[556]:
```
df_final.plot.scatter(x="Metascore", y="rating")
```

[556]: <Axes: xlabel='Metascore', ylabel='rating'>



[557]:
```
df_final.groupby("Genres_x")["rating"].mean().sort_values(ascending=False).
↪head(10)
```

[557]:
```
Genres_x
Single-player,Captions available,Steam Cloud
0.909091
Single-player,Custom Volume Controls,Mouse Only Option,Playable without Timed
Input,Save Anytime,Partial Controller Support    0.740741
Single-player,VR Only,Partial Controller Support,Family Sharing
0.129213
Single-player,Steam Trading Cards,Partial Controller Support
0.059905
Single-player,Multi-player,PvP,Online PvP,Co-op,Online Co-op,Full controller
support                                           0.058157
Single-player,Steam Achievements,In-App Purchases,Steam Cloud
0.039337
Single-player,Steam Achievements,Steam Trading Cards,Steam Leaderboards,Family
Sharing                                          0.017652
Single-player,Multi-player,PvP,Online PvP,Co-op,Online Co-op,Steam
```

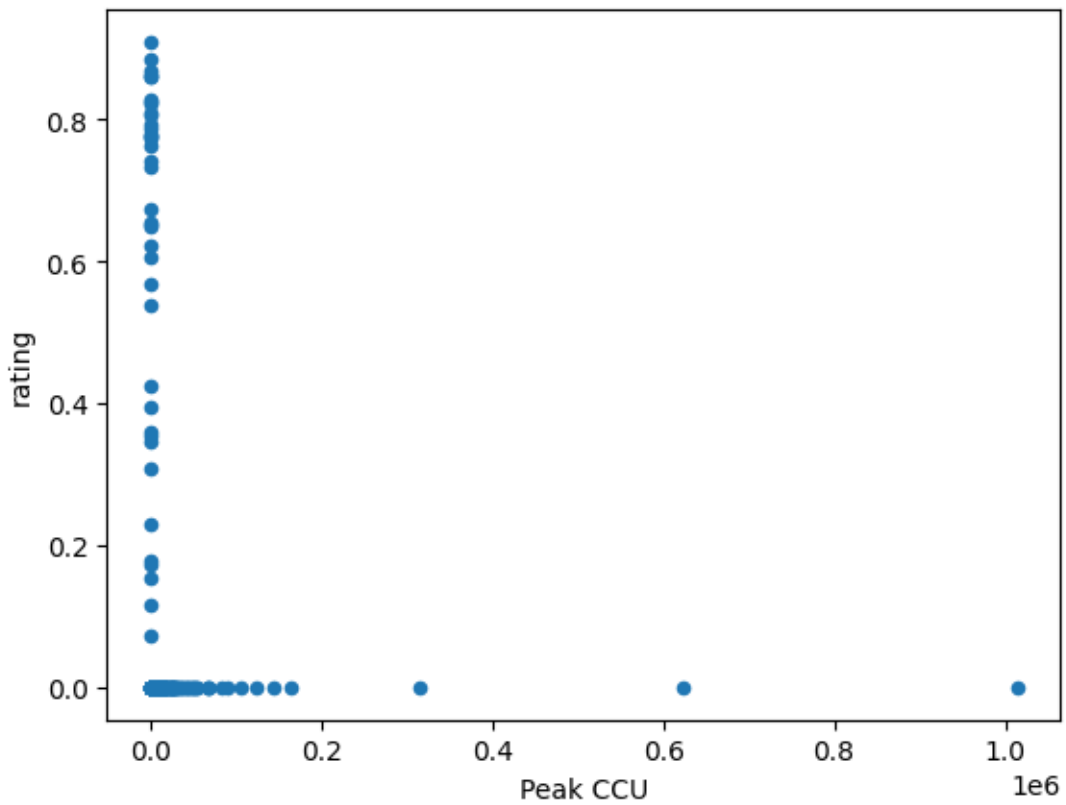
```
Achievements,Steam Trading Cards,Family Sharing              0.013340
Multi-player,MMO,PvP,Online PvP,Co-op,Online Co-op,Family Sharing
0.012333
Single-player,Steam Achievements,Stats,Steam Leaderboards,Family Sharing
0.003768
Name: rating, dtype: float64
```

[558]:
```
df_final.groupby(df_final["Price"] == 0)["Recommendations"].mean()
```

[558]:
```
Price
False    20.918251
True      7.671933
Name: Recommendations, dtype: float64
```

[559]:
```
df_final.plot.scatter(x="Peak CCU", y="rating")
```

[559]:
```
<Axes: xlabel='Peak CCU', ylabel='rating'>
```



[560]:
```
df_final.groupby("year")["rating"].mean()
```

[560]:
```
year
1997    0.000000
1998    0.000000
1999    0.000000
2000    0.000000
2001    0.000000
2002    0.000000
2003    0.000000
2004    0.000000
2005    0.000000
2006    0.000000
2007    0.000000
2008    0.000000
2009    0.000000
2010    0.000000
2011    0.000000
2012    0.000000
2013    0.000000
2014    0.000000
2015    0.000124
2016    0.000574
2017    0.000683
2018    0.002010
2019    0.000549
2020    0.000022
2021    0.000000
2022    0.000000
2023    0.000000
2024    0.000000
2025    0.000000
2026         NaN
Name: rating, dtype: float64
```